

Coding & Solution

Date	03 October 2026
Team ID	SWTID-2026-5999
Project Name	PocketSmart AI — AI-Assisted Goal-Based Budget Planner
Maximum Marks	5 Marks

Solution Summary

Field	Details
Repository	https://github.com/KAWIN-MURUKANANTHAM/Pocket-Smart-AI
Languages	Python, JavaScript, HTML, CSS
Frameworks	FastAPI, Uvicorn, Jinja2, Pydantic
AI Integration	Google Gemini through <code>google-genai</code>
Core Implementation	Budget-first planners, server-side calculation, shopping search links, accounts, history and fallback handling
Current Limitations	External links are search links; prices are estimates; no direct checkout or live retailer inventory
Run Method	Install dependencies, configure environment variables and start the FastAPI application with Uvicorn

Implementation Approach

1. **Validate the request** using Pydantic models before the AI call.
2. **Build a planner-specific prompt** from the validated budget and requirements.
3. **Request structured output** from Gemini.
4. **Sanitize the response** and convert prices/quantities into safe numeric values.
5. **Recalculate totals on the server** rather than trusting the model’s arithmetic.
6. **Trim items when required** to keep the plan within the submitted limit.
7. **Attach search links** according to the recommendation category.
8. **Save the completed plan** to the user’s history.

Code Quality Checklist

S.No	Criteria	Status
1	Clear module separation	Yes
2	Meaningful names and validation rules	Yes
3	Error handling around authentication, uploads and AI calls	Yes
4	Reusable budget and shopping-link helpers	Yes
5	Environment-based configuration for secrets	Yes
6	Automated tests included	Yes

Important Engineering Decision

The project does not allow an AI response to directly decide the final amount. The backend owns the calculation and budget constraint, which makes the application behaviour more predictable.